Measurement of Absolute Double Differential Cross-Section in Invariant Mass and p_T Bins

Chatura Kuruppu

SeaQuest Experiment (E906)
Fermi National Accelerator Laboratory

May 7, 2026

Input Files Used

Note: We do not use RS5a and RS5b files for this study.

ROOT files used:

- merged_RS57_Empty_2_1138_FinalData.root
- merged_RS57_LD2_3_1138_FinalData.root
- merged_RS57_LH2_1_1138_FinalData.root
- merged_RS59_Empty_2_466_FinalData.root
- merged_RS59_LD2_3_466_FinalData.root
- merged_RS59_LH2_1_465_FinalData.root
- merged_RS5a_Empty_2_1680_FinalData.root
- merged_RS5a_LD2_3_1689_FinalData.root
- merged_RS5a_LH2_1_1476_FinalData.root
- merged_RS5b_Empty_2_917_FinalData.root
- merged_RS5b_LD2_3_918_FinalData.root
- merged_RS5b_LH2_1_770_FinalData.root
- merged_RS62_Empty_2_1234_FinalData.root
- merged_RS62_LD2_3_1237_FinalData.root
- merged_RS62_LH2_1_1234_FinalData.root
- merged_RS67_Empty_2_14_FinalData.root
- merged_RS67_LD2_3_15_FinalData.root
- merged_RS67_LH2_1_5_FinalData.root
- merged_RS70_Empty_2_267_FinalData.root
- merged_RS70_LD2_3_266_FinalData.root
- merged_RS70_LH2_1_264_FinalData.root

Kinematic Phase Space

Kinematic Binning Definition:

- **Mass Bins (GeV):** [4.2, 4.5, 4.8, 5.1, 5.4, 5.7, 6.0, 6.3, 6.6, 6.9, 7.5, 8.7]
- p_T **Bins (GeV):** [0.0, 0.32, 0.49, 0.63, 0.77, 0.95, 1.18, 1.8]
- x_F **Bins:** [0.0, 0.85] with a fixed step width of 0.05.

Event Selection Criteria

Dimuon Kinematics:

- $4.2 < \text{Mass} < 8.8$ GeV, $-0.1 < x_F < 0.95$, $0.05 < x_T \leq 0.58$, and $|\cos\theta| < 0.5$.
- Vertex cuts: $-280 < dz < -5$, target/dump tracking separation criteria met.
- Transverse momentum bounds: $|dp_x| < 1.8$, $|dp_y| < 2.0$, and $dp_x^2 + dp_y^2 < 5$.

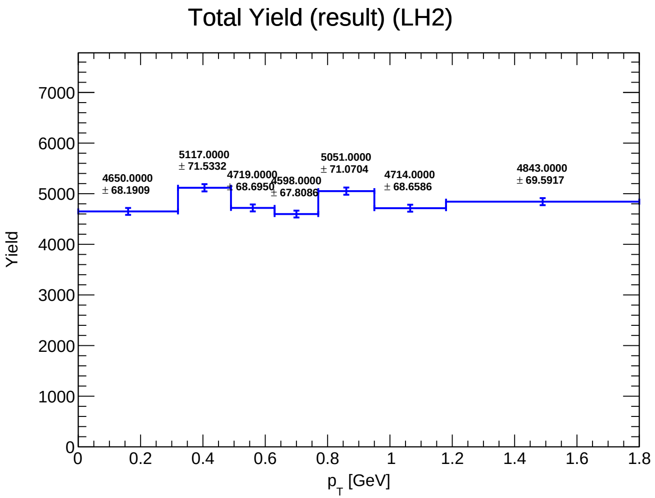
Track Quality & Acceptance:

- $\chi^2_{\text{target}} < 15$, $\chi^2_{\text{dimuon}} < 18$, and $\chi^2/(\text{nHits} - 5) < 12$.
- Hit requirements: $\text{nHits} > 13$ per track, $\text{nHits}_1 + \text{nHits}_2 > 29$, St1 total > 8.
- Track vertex longitudinal limits: $-320 < z_v < -5$.
- A dynamic beam offset correction is applied to y-coordinates: offset is 1.6 for runID ≥ 11000 and 0.4 otherwise.

Detector Occupancy Limits:

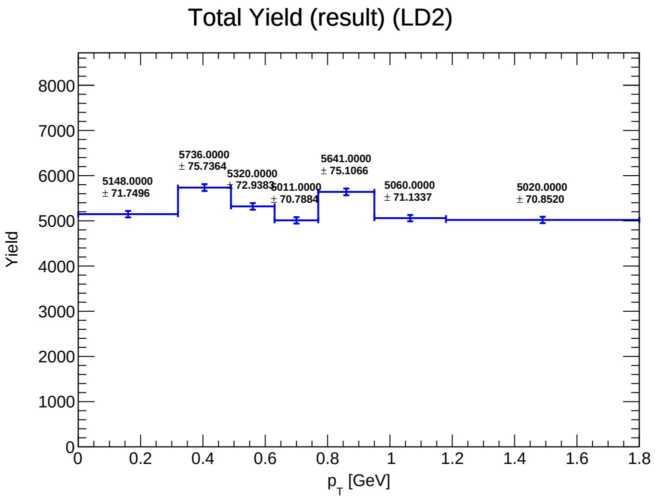
- Drift chamber hits: $D1 < 400$, $D2 < 400$, $D3 < 400$, and total $D1 + D2 + D3 < 1000$.

Y_total_LH2

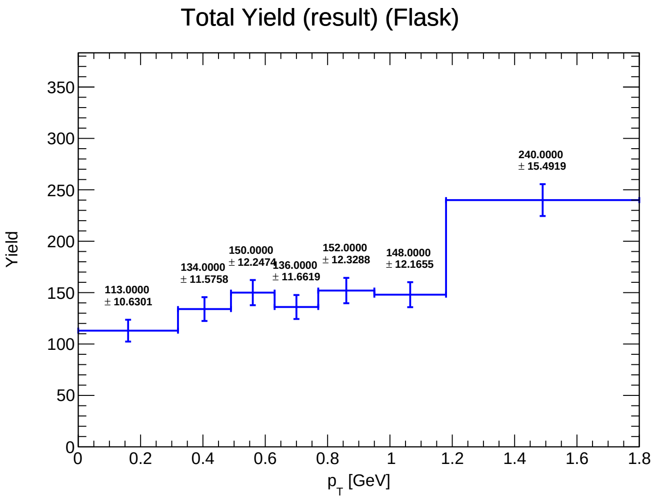


Total Yield (result) (LH2)
Yield
p_T [GeV]
4650.0000
± 68.1909
5117.0000
± 71.5332
4719.0000
± 68.6950
4598.0000
± 67.8086
5051.0000
± 71.0704
4714.0000
± 68.6586
4843.0000
± 69.5917
0
1000
2000
3000
4000
5000
6000
7000
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8

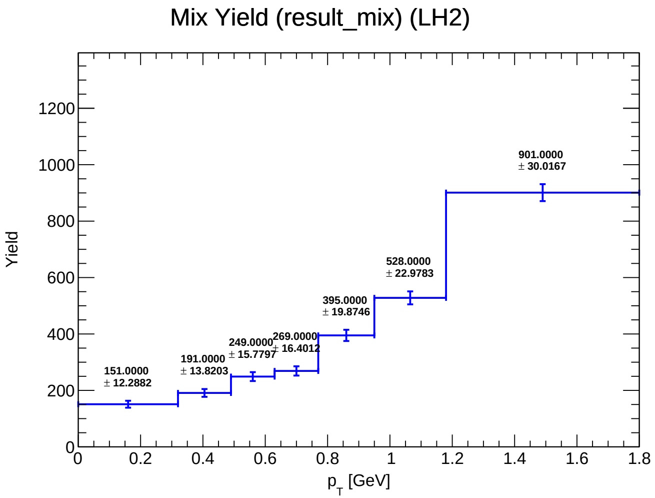
Y_total_LD2


Total Yield (result) (LD2)
5148.0000
± 71.7496
5736.0000
± 75.7364
5320.0000
± 72.9383
5011.0000
± 70.7884
5641.0000
± 75.1066
5060.0000
± 71.1337
5020.0000
± 70.8520
Yield
0
1000
2000
3000
4000
5000
6000
7000
8000
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
p_T [GeV]

Y_total_Flask

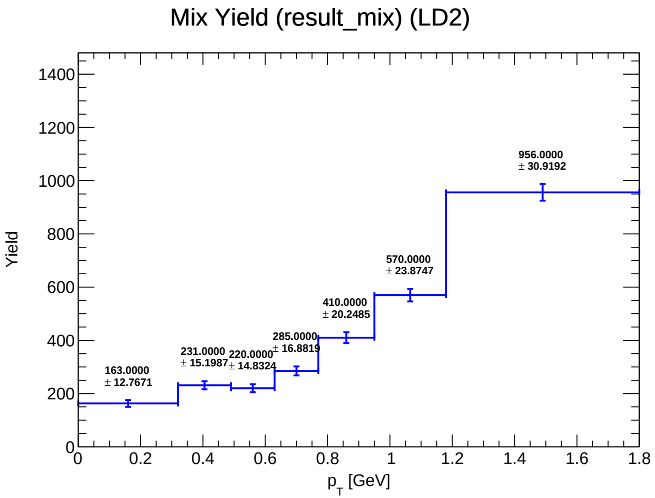


Y_mix_LH2

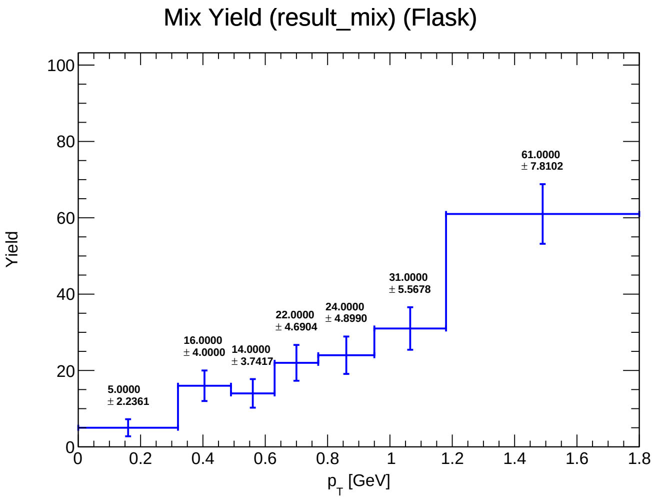


Mix Yield (result_mix) (LH2)
Yield
p_T [GeV]
0
200
400
600
800
1000
1200
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
151.0000
± 12.2882
191.0000
± 13.8203
249.0000
± 15.7797
269.0000
± 16.4012
395.0000
± 19.8746
528.0000
± 22.9783
901.0000
± 30.0167

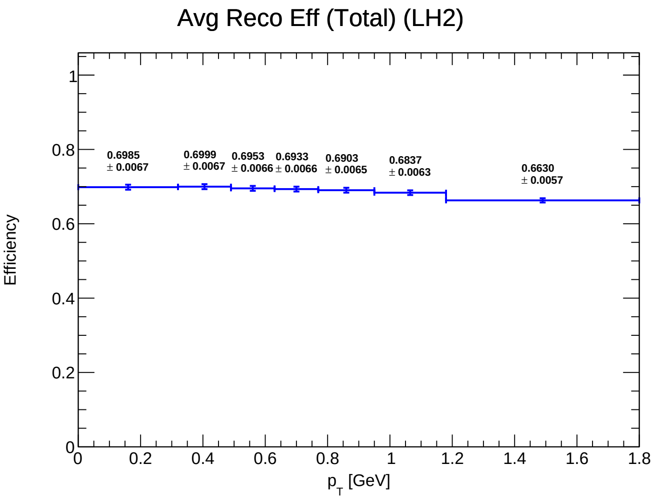
Y_mix_LD2


Mix Yield (result_mix) (LD2)
Yield
p_T [GeV]
0
200
400
600
800
1000
1200
1400
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
163.0000
± 12.7671
231.0000
± 15.1987
220.0000
± 14.8324
285.0000
± 16.8819
410.0000
± 20.2485
570.0000
± 23.8747
956.0000
± 30.9192

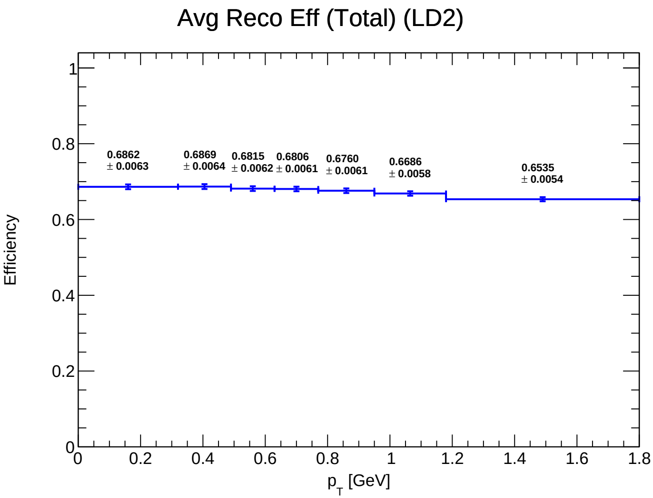
Y_mix_Flask


Mix Yield (result_mix) (Flask)
Yield
p_T [GeV]
0
20
40
60
80
100
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
5.0000
± 2.2361
16.0000
± 4.0000
14.0000
± 3.7417
22.0000
± 4.6904
24.0000
± 4.8990
31.0000
± 5.5678
61.0000
± 7.8102

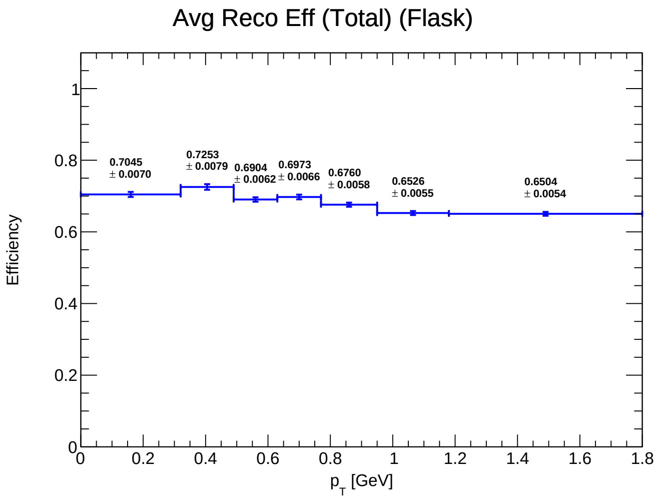
E_total_reco_LH2



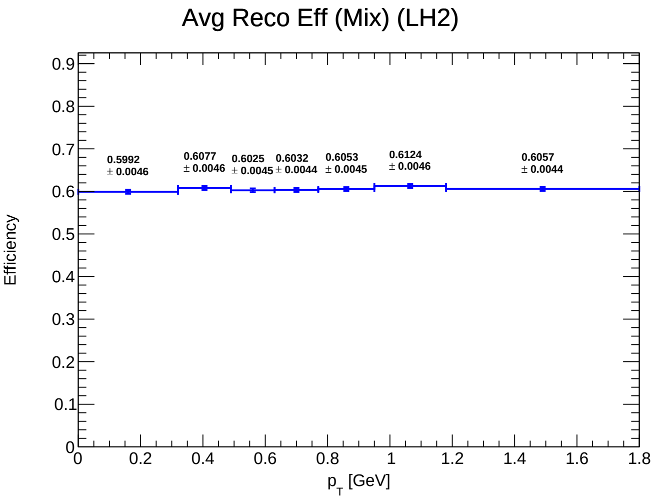
E_total_reco_LD2



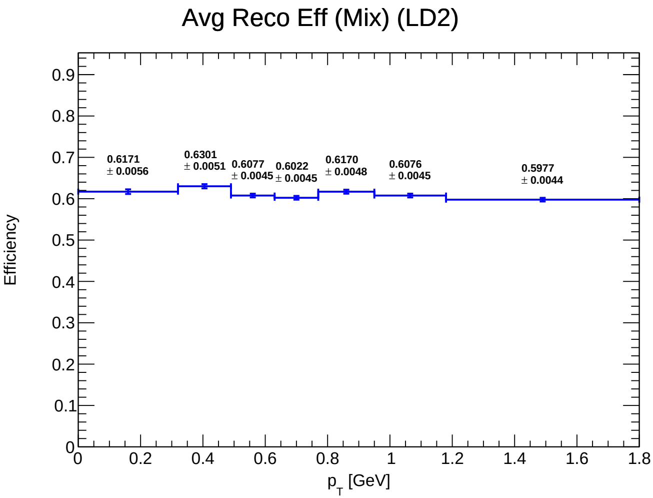
E_total_reco_Flask



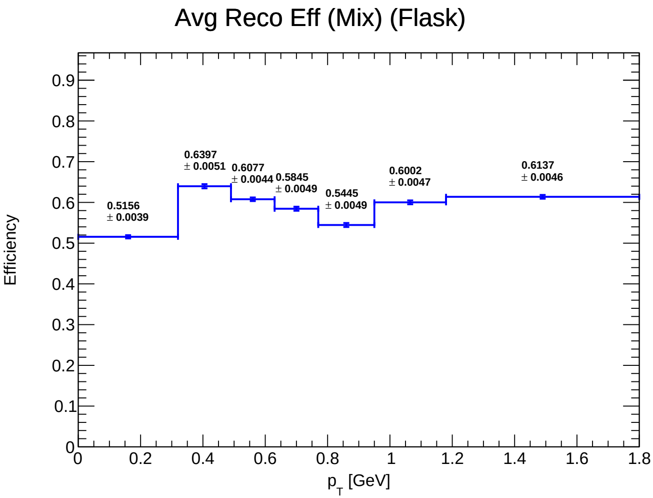
E_mix_reco_LH2



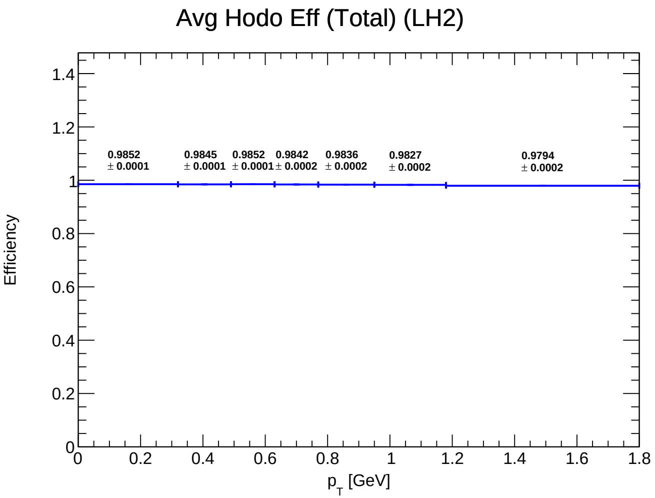
E_mix_reco_LD2



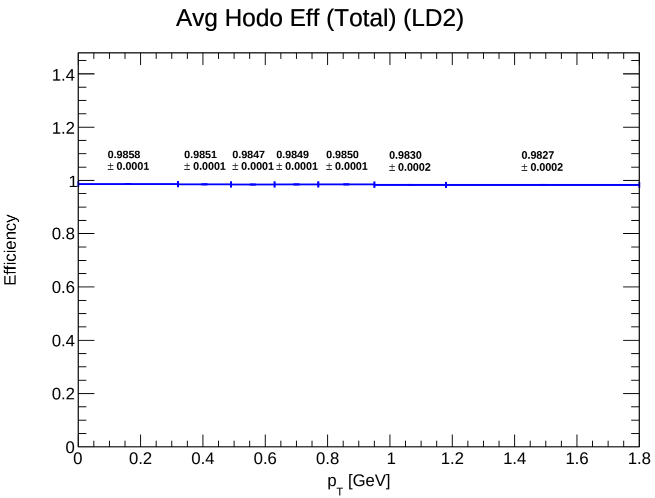
E_mix_reco_Flask



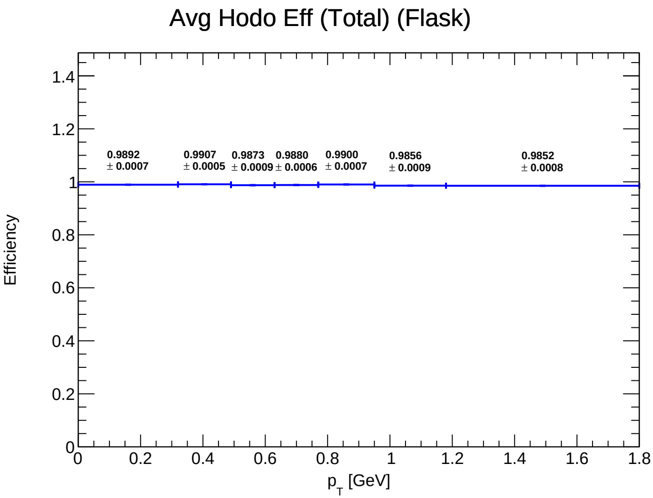
E_total_hodo_LH2



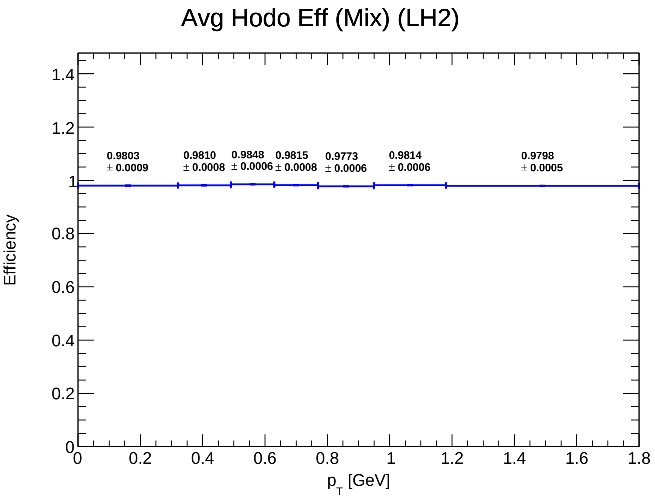
E_total_hodo_LD2



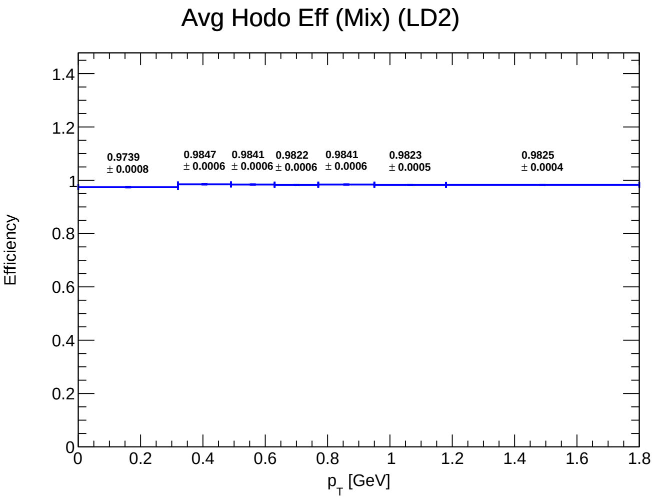
E_total_hodo_Flask



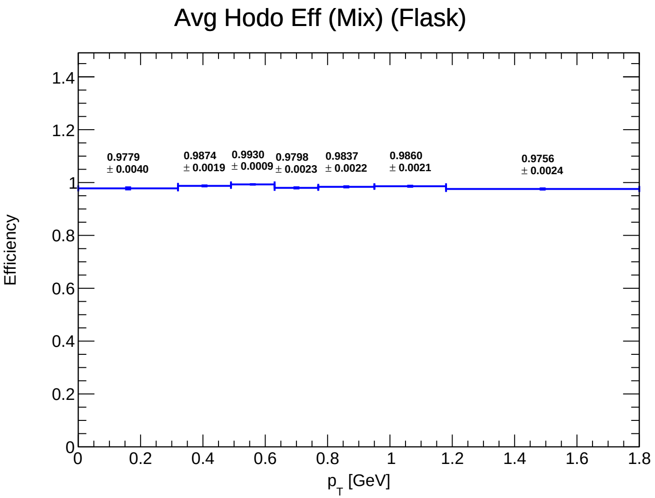
E_mix_hodo_LH2



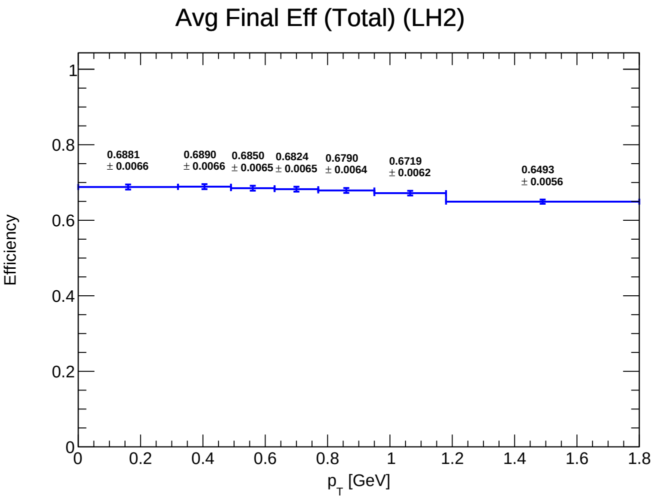
E_mix_hodo_LD2



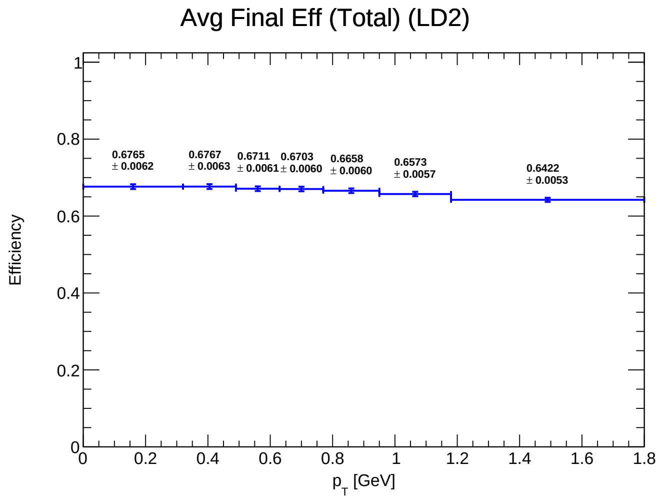
E_mix_hodo_Flask



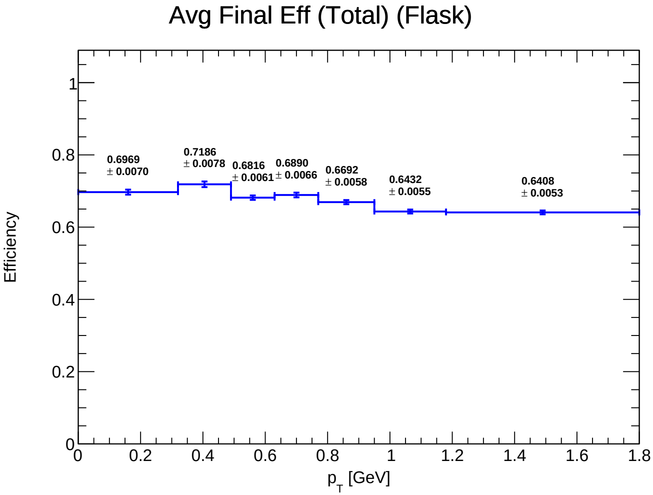
E_total_final_LH2



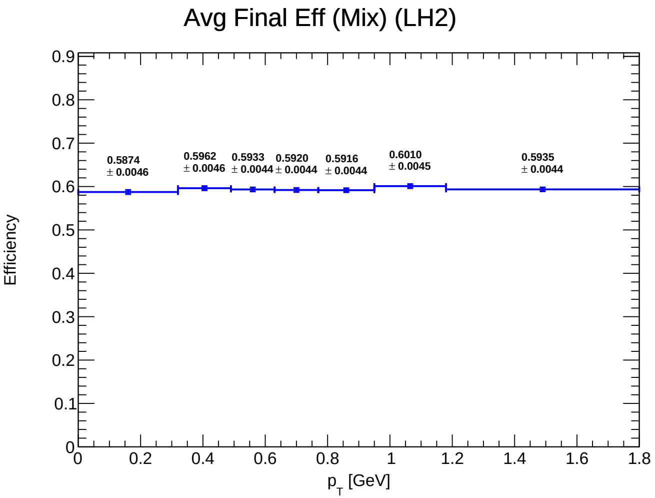
E_total_final_LD2



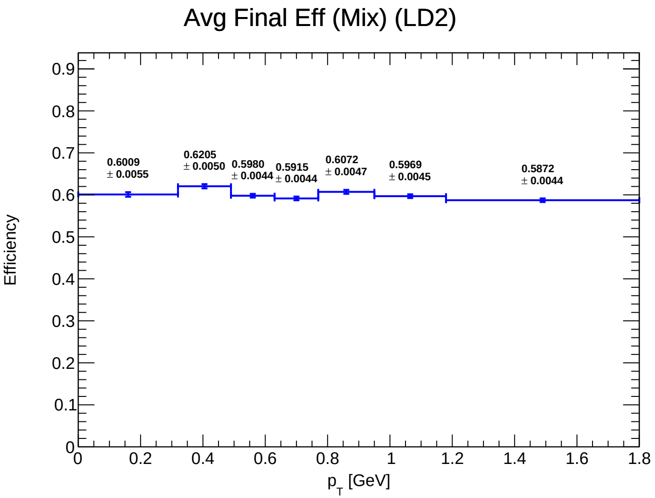
E_total_final_Flask



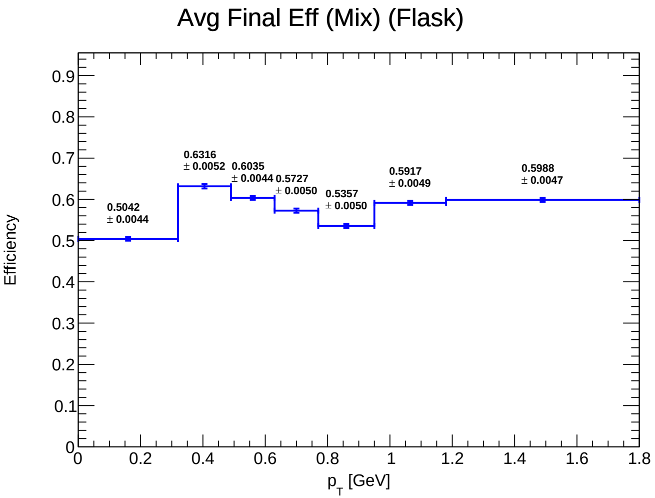
E_mix_final_LH2



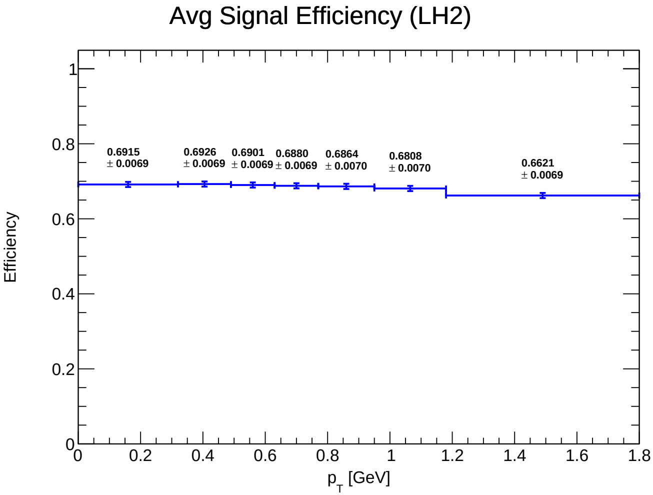
E_mix_final_LD2



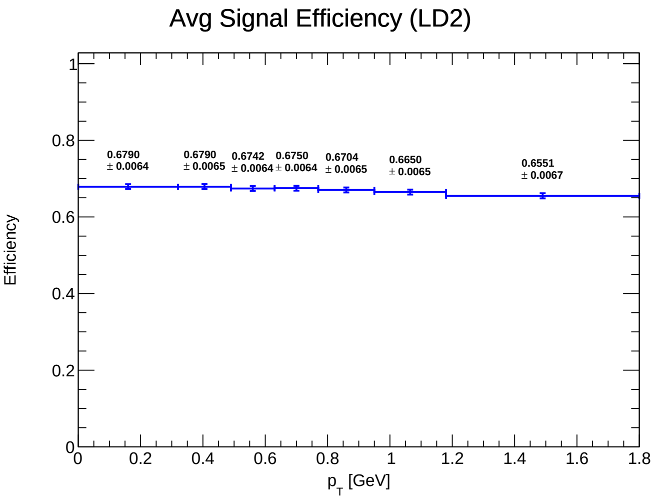
E_mix_final_Flask



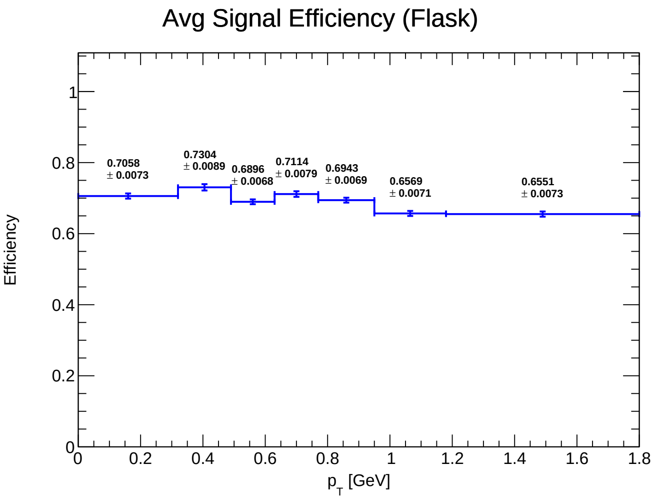
E_final_signal_LH2



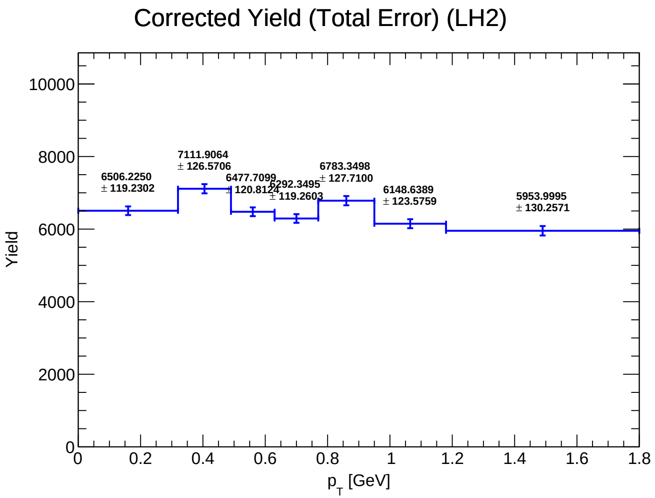
E_final_signal_LD2



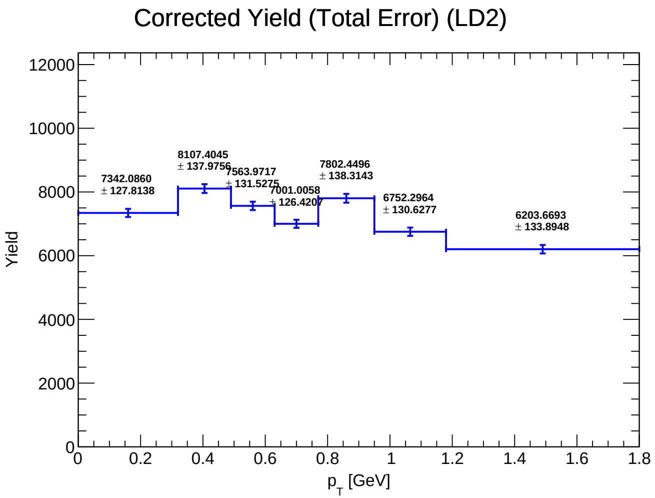
E_final_signal_Flask



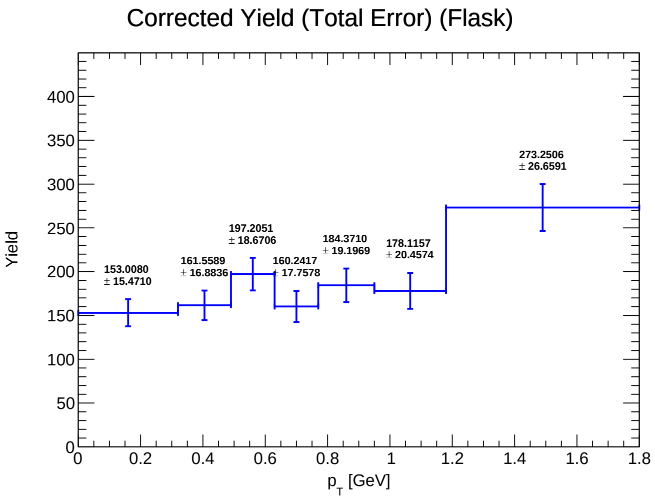
Y_corrected_LH2

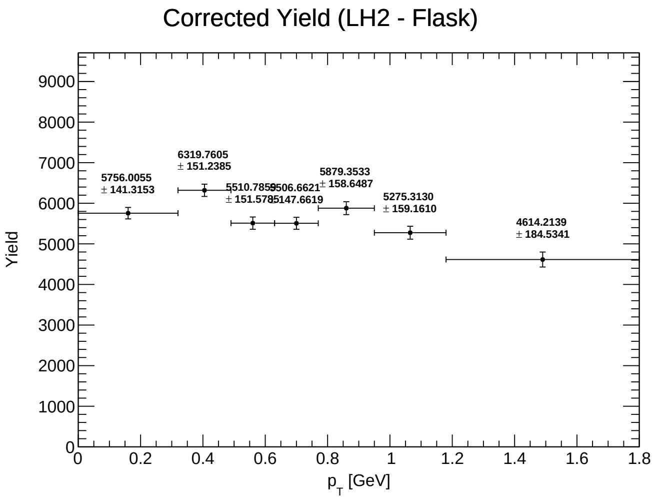


Y_corrected_LD2

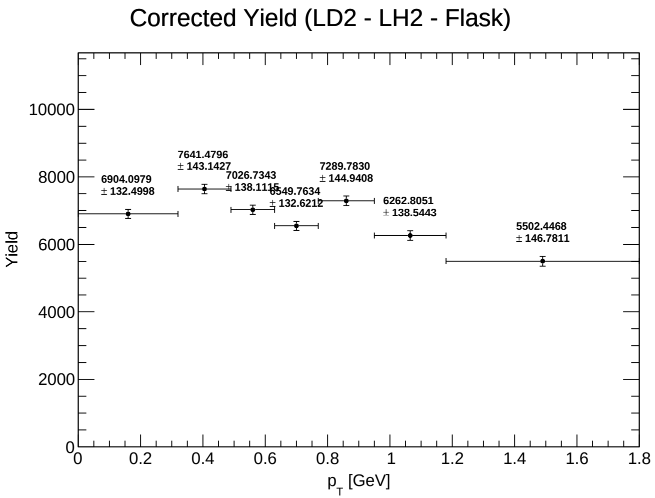


Y_corrected_Flask

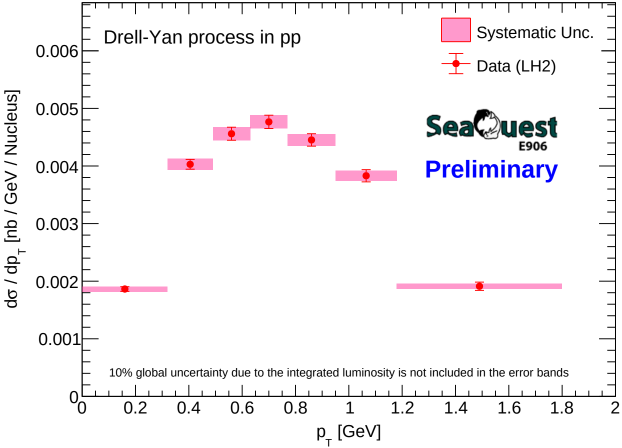



Corrected Yield (LH2 - Flask)
Yield
p_T [GeV]
5756.0055
± 141.3153
6319.7605
± 151.2385
5510.7859
± 151.5785
5506.6621
± 147.6619
5879.3533
± 158.6487
5275.3130
± 159.1610
4614.2139
± 184.5341
0
1000
2000
3000
4000
5000
6000
7000
8000
9000
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8

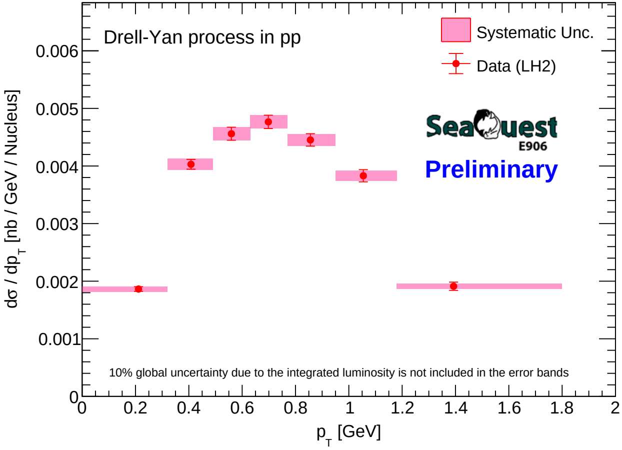
Y_corrected_Subtracted_LD2



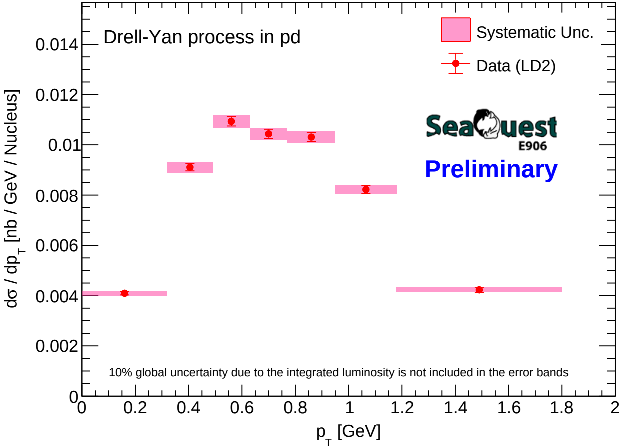
CrossSection_LH2_geom_vs_pT_with_logo



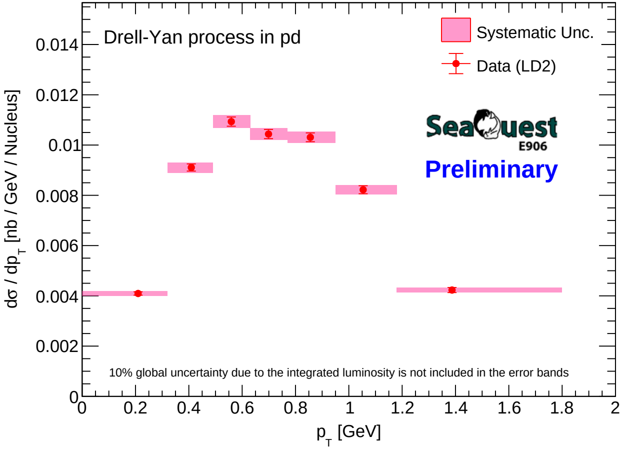
CrossSection_LH2_true_pt_vs_pT_with_logo



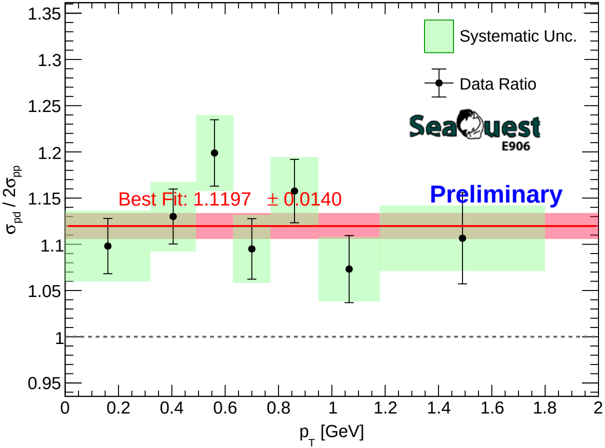
CrossSection_LD2_geom_vs_pT_with_logo



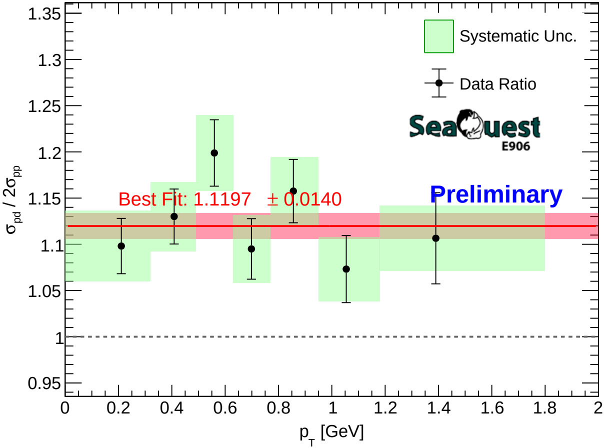
CrossSection_LD2_true_pt_vs_pT_with_logo



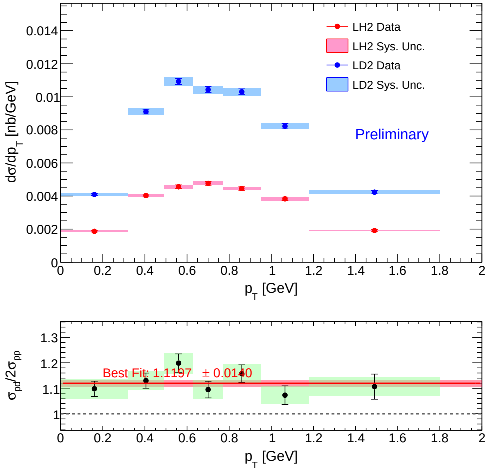
CrossSection_Ratio_pd_2pp_vs_pT_geom_with_logo



CrossSection_Ratio_pd_2pp_vs_pT_true_pt_with_logo



Combined_XSec_Ratio_vs_pT_geom



Combined_XSec_Ratio_vs_pT_true_pt